Dynamic Downsampling

Tanya Strydom [1]; Andrew P. Beckerman [1]

Abstract: TODO

Keywords: food webs

1 Introduction

Extinction simulation in networks are useful because they can inform us as to the robustness [REF] or resilience [REF] of the system. Understanding community level extinctions is of particular interest for understanding community collapse and mass extinction events [REFs]. However the nature of the fossil record means that we are only able to do topological extinctions [REFs]

Dynamic extinctions (*e.g.,* Curtsdotter et al., 2011) can be useful because they are (probably) a better approximation of reality - or at least not as 'conservative'/best case scenario as what the topological extinctions are (insert a brief description of how these approaches are different). Something about how this is meaningful in paleo but also real-world settings. Maybe also point out that typically in dynamic context we are using species agnostic models *e.g.,* niche and atn which lack the real world tractability that we want.

However in order to *correctly* match the philosophy of these dynamic networks we need a way in which to turn empirical metawebs into networks that are more representative of realised networks, *i.e.,* there is some form of link distribution constraint (Strydom, Dunhill, et al., 2026). Although there has been some work on downsampling of paleo webs (Roopnarine, 2006) we lack any ground truthing of what the implications are of embedding these ecological assumptions. Additionally this downsampling approach makes insidious assumptions as to what is driving the distribution of links in a network. Possibly also add a sentence here about how the other work has focused on the niche model (Curtsdotter et al., 2011; Jonsson et al., 2015) which itself has its own sets of ecological assumptions/embeddings.

Here we present a selection of downsampling approach that are built on different assumptions as to link constraints. Using these downsampling approaches we recreate the Curtsdotter et al. (2011) approach and test if we see difference in topological and dynamic extinctions, as well as if the different input networks have different signals. We do this using the same dataset from Karapunar et al. (2026). If realised webs emerge through different ecological filters, do our conclusions about dynamic extinctions depend on which realisation process we assume?

2 Methods

2.1 Data

Seven locations (Karapunar et al., 2026). Here we use the guild-level community data to keep the size of the networks manageable and well within the bounds used by Curtsdotter et al. (2011). Additionally we added plankton and zooplankton data from EcoGenie models [REF] as a way to enlarge the 'base' of the network (again to better represent a realised network).

2.2 Network construction

For the empirical networks we used the PFWIM model (Shaw et al., 2024). *Insert brief description about trait assignments and mechanistic assumptions, mention we wanted to go as broad as possible so we used fully categorical rules.* This was then used as the base metaweb for each community. Each metaweb was then downsampled to a target connectance (see below for the different downsampling approaches). Additionally the target connectance and species richness was used to parameterise a niche model network (Williams & Martinez, 2000). We also constructed ATN networks (Brose et al., 2006).

2.3 Downsampling approaches

We treat downsampling as a process of **ecological realisation** rather than observation. Starting from a metaweb containing all potential trophic interactions, each downsampling approach represents an alternative ecological hypothesis describing how local ecological processes filter these potential interactions into the subset that is realised within a particular community. The different approaches therefore differ not only in their implementation, but also in the ecological assumptions they embed about how realised food webs emerge from a common regional interaction pool.

2.3.1 Niche

The original niche model from Williams & Martinez (2000) assumes that realised interactions represent the core of a consumer's trophic niche, and that these diets can be arranged along a single niche axis. In order to create a downsampling approach that mimics this we we infer a latent trophic axis directly from the metaweb using Singular Value Decomposition (SVD). It has been shown that the latent spaces described by the left and right singular vectors are closely aligned wth the generality and vulnerability of species (Strydom et al., 2022) and so the SVD composition should retain the diet of the species.

Let A denote the $S \times S$ adjacency matrix of the metaweb, where $A_{ij} = 1$ if consumer i feeds on resource j. We compute its singular value decomposition,

$$A = U \Sigma V^{\top},$$

where U and V are the left and right singular vectors, respectively, and Σ is the diagonal matrix of singular values. Using the leading singular value, σ_1, and corresponding singular vectors, $\mathbf{u}_1$ and $\mathbf{v}_1$, each species is assigned latent consumer and resource coordinates,

$$c_i = \sqrt{\sigma_1}\, u_{i1}, \qquad r_i = \sqrt{\sigma_1}\, v_{i1}.$$

Because species can act both as consumers and as resources, these coordinates are combined to obtain a single niche position,

$$x_i = \frac{c_i + r_i}{2},$$

which is subsequently rescaled to lie between 0 and 1. Species with similar values of x_i therefore occupy similar positions along the dominant trophic gradient of the metaweb.

For each consumer i, the centroid of its realised niche is

$$\mu_i = \frac{1}{k_i} \sum_{j \in R_i} x_j,$$

where R_i is the set of resources consumed by species i, and k_i is its trophic breadth (generality). Niche breadth is estimated as the standard deviation of prey positions,

$$\sigma_i = \max\left(\mathrm{sd}(x_j),\, 0.1\right),$$

where the lower bound prevents unrealistically narrow niches for specialists. Niche breadth can optionally be scaled to further constrain realised diets,

$$\sigma_i' = \alpha \sigma_i,$$

where α is acts as a scaling parameter.

The probability of retaining an interaction between consumer i and resource j is then

$$P_{ij} \propto \exp\left[-\left(\frac{x_j - \mu_i}{\sigma_i'}\right)^2\right].$$

Interactions closest to the consumer's niche centre are therefore most likely to be realised (retained), whereas interactions towards the margins of the niche are preferentially removed. Ecologically, this assumes that local communities realise the central portion of a consumer's potential trophic niche, with peripheral or opportunistic interactions occurring less consistently.

2.3.2 Power-law

The power-law approach assumes that interaction realisation depends primarily on **consumer trophic breadth**. Following Roopnarine (2006), the retention probability for consumer i is

$$P_i \propto \exp\left(\frac{k_i}{E}\right),$$

where k_i is consumer generality and

$$E = S^{(y-1)/y},$$

with S denoting network size and y the scaling exponent. Probabilities are subsequently normalised to lie between 0 and 1. Every interaction belonging to consumer i inherits the same retention probability,

$$P_{ij} = P_i.$$

Ecologically, this represents the hypothesis that generalist consumers realise a larger fraction of their potential trophic niche than specialist consumers, irrespective of the identity of individual resources.

2.3.3 Degree-product

The degree-product approach assumes that interaction realisation depends on the centrality of both interacting species. Each interaction is assigned a weight

$$w_{ij} = k_i^{\text{out}} k_j^{\text{in}},$$

where k_i^{out} is the consumer's out-degree (generality) and k_j^{in} is the resource's in-degree (vulnerability). Retention probabilities are then calculated as

$$P_{ij} = 1 - \frac{w_{ij}}{\max(w)},$$

before being constrained to the interval $[0,1]$. Under iterative downsampling, degrees are recalculated following each interaction removal, allowing probabilities to evolve as the realised network changes. Ecologically, this assumes that highly connected species possess many potential interactions that are not simultaneously realised within any local community, whereas interactions involving more specialised or weakly connected species are more likely to represent essential realised links.

2.3.4 Random

Random downsampling provides the ecological null model. Every interaction has an equal probability of being retained,

$$P_{ij} = c,$$

where c is constant for all existing interactions. Interactions are removed uniformly at random until the target connectance is reached, subject to the constraint that species cannot become isolated. This assumes that no ecological process preferentially favours the realisation of one potential interaction over another, such that realised food webs constitute an unbiased subset of the regional metaweb that is ultimately only constrained by the number of links.

2.4 Simulations

For each empirical community, we generated 30 replicate network ensembles. Species body masses were sampled independently from truncated log-normal distributions constrained by their categorical body-size classifications. Initial biomasses were drawn from a uniform distribution. These trait values were used to construct a series of alternative networks, representing different two realised networks (ATN and niche model), one metaweb, and four downsampled (realised) metawebs (niche, power-law, degree-product, and random).

For the allometric trophic network (ATN), interaction strengths were generated by varying the feeding threshold between 0.01 and 0.12 and selecting the network whose connectance most closely matched a target connectance. Target connectance was independently sampled for each replicate from a truncated normal distribution bounded between 0.05 and 0.15. The target connectance and richness of the metaweb were used to build a niche model and the metaweb was downsampled until the target connectance was reach. These represent the ‘creation’ networks.

Each creation network was then parameterised using the END package (Delmas et al., 2017) and simulated for up to 5000 time steps. Simulations terminated early if a dynamical equilibrium was reached. Species whose biomass declined below the survival threshold (10^{-12}) during this burn-in period were considered extinct and removed from the network. The surviving species and their interactions constituted the corresponding 'realised' networks.

Extinction experiments were subsequently performed on both the creation and realised networks. Topological extinction simulations were conducted on both network stages, whereas dynamic extinction simulations were performed only on realised networks. In all simulations, primary extinctions followed one of the prescribed extinction scenarios. During topological simulations, secondary extinctions occurred when species lost all prey, these were allowed to cascade. During dynamic simulations, primary extinctions were imposed by setting the biomass of the focal species to the survival threshold (10^{-12}), after which the community was simulated for a further 5000 time steps to allow biomass dynamics to relax. Species whose biomass subsequently declined below the survival threshold were recorded as secondary extinctions. This procedure was repeated until the entire community collapsed.

For each extinction scenario, robustness was quantified from the resulting extinction curves as the proportion of primary extinctions required to produce a 50% reduction in species richness (R_{50} (Jonsson et al., 2015)), estimated by linear interpolation between extinction events.

2.5 Software

Network construction and extinction simulations were run in Julia v1.12.6 (Bezanson et al., 2017), we used `EcologicalNetworksDynamics.jl` (Lajaaiti et al., 2025) for dynamic simulations, `PFIM.jl` [REF] for construction of metawebs, `FoodWebTools.jl` [REF] for network downsampling, and Extinctions.jl [REF] for extinctions and robustness calculations. Statistical analyses were run in R v4.5.2 [REF]. All code can be found at *TODO*

> Keep the narrative extinction centric for now but remember we also have the topology of each network before the various extinction simulations begin

3 Downsampling and burn-in alter network topology in method-dependent ways

Redundancy analysis (RDA), using permutation-based significance testing while accounting for community-level variation, revealed that downsampling approach significantly affected network topology, but that this effect depended on the magnitude of connectance reduction required to reach the target connectance (999 permutations; $F_{7,707} = 112.8$, $p = 0.001$; Figure 1, panel A). The model explained 31.8% of total topology variation (adjusted $R^2 = 0.318$), with community identity accounting for an additional 39.7% of variation. The significant interaction between downsampling approach and connectance change (marginal test; $F_{3,707} = 18.90$, $p = 0.001$) demonstrated that topology trajectories differed among downsampling methods as connectance declined. Thus, the extent to which network structure diverged from the original topology depended on both the severity of downsampling and the mechanism by which interactions were removed.

[Figure 1 about here.]

To identify the source of this interaction, the response of the primary RDA axis was examined across the connectance gradient. RDA1 was strongly associated with connectance change ($F_{1,712} = 2076.86$, $p < 0.001$), with additional effects of downsampling approach ($F_{3,712} = 33.28$, $p < 0.001$) and a significant downsampling approach × connectance change interaction ($F_{3,712} = 4.81$, $p = 0.003$). Pairwise comparisons of estimated trajectory slopes showed that link-based downsampling followed a distinct trajectory, differing from niche-, power-, and random-based downsampling, whereas the latter three approaches exhibited statistically indistinguishable responses (Figure 1, panel B). These results indicate that all downsampling methods alter network topology as connectance declines, but that removing interactions according to their realised connectivity produces a fundamentally different pathway of structural change from approaches that incorporate species-level ecological information or random link loss.

We next asked whether these structurally distinct networks responded differently during dynamic burn-in. Structural reorganisation was quantified as the Euclidean displacement between network positions before and after burn-in in the first three principal component dimensions, with community included as a random effect to account for the substantial variation among communities. Burn-in displacement differed significantly among network types ($F = 62.0$, $p < 0.001$). Metawebs underwent the greatest structural reorganisation, whereas ATNs exhibited the smallest displacement. Among the downsampled networks, niche-, power-, and link-based approaches underwent comparable structural change during burn-in, while random downsampling showed a modestly greater displacement (Figure 1, panels C–D).

Together, these analyses demonstrate that downsampling approach influences network topology both during network construction and subsequent dynamic relaxation. Differences among downsampling methods become increasingly pronounced as greater proportions of interactions are removed, consistent with the expectation that the ecological assumptions underlying each removal strategy exert a stronger influence under more aggressive downsampling. Importantly, despite generating distinct initial topologies, the different downsampling approaches underwent remarkably similar structural reorganisation during burn-in, changing in ways more similar to the niche model than the original metaweb. In contrast, random downsampling remained distinct throughout burn-in, consistent with its lack of an underlying ecological mechanism governing interaction removal. This suggests that the ecologically informed downsampling on metawebs allows us to produce networks whose subsequent structural dynamics more closely resemble those of the niche model. Notably the two realised models (Niche and ATN) do not behave similarly and still end up at different 'end points' post burn-in to the empirical webs, highlighting that network reconstruction approach will still have a strong effect on the inferred dynamics and behaviour of the system (Strydom, Karapunar, et al., 2026)

4 Differences in inferred robustness of topological and dynamic are unaffected by network type

To determine whether the qualitative differences between topological and dynamic extinction simulations previously demonstrated using niche-model food webs generalise across alternative network reconstructions, we compared the difference in robustness ($\Delta R_5 0$) among network types for each extinction scenario while accounting for community-level variation. Network type significantly affected $\Delta R_5 0$ in all four extinction scenarios (low-to-high biomass: $F_{6,1248} = 22.3$, p<0.001; high-to-low biomass: $F_{6,1248} = 46.7$, p<0.001; random basal: $F_{6,1248} = 109.0$, p<0.001; random consumer: $F_{6,1248} = 22.9$, p<0.001), indicating that the magnitude of the difference between topological and dynamic robustness depended on network reconstruction.

Despite these quantitative differences, the qualitative pattern identified by Curtsdotter et al. (2011) was largely maintained. Under all but random basal removal, topological extinction simulations inferred greater robustness than dynamic simulations for every network type, although the magnitude of this discrepancy varied among reconstruction methods Figure 2. Random basal extinction represented the only notable departure from this general pattern. Here, the discrepancy between topological and dynamic robustness was substantially reduced, with link- and power-downsampled networks exhibiting slightly positive values of $\Delta R_5 0$ and random downsampled networks showing little difference between the two approaches, whereas niche-derived networks retained lower dynamic than topological robustness. Thus, the conclusion that topological analyses provide more conservative estimates of robustness was not universal, but remained remarkably robust across reconstruction methods and extinction scenarios.

[Figure 2 about here.]

Together, these analyses demonstrate that the principal conclusion of Curtsdotter et al. (2011) extends well beyond the niche model. While the precise magnitude of the discrepancy between topological and dynamic robustness depends on both network reconstruction and extinction mechanism, the qualitative tendency for topological extinction simulations to infer greater robustness than dynamic simulations was recovered across the vast majority of reconstructed food webs examined here.

5 Downsampling and burn-in alter inferred topological robustness

this is probably destined as a supp matt exercise

To determine how ecological assumptions underlying downsampling influence emergent robustness, we examined how differences in robustness between downsampled networks and three reference 'states' (metaweb, ATN, and niche model) changed with increasing degree of downsampling ($\Delta_{connectance}$). As the degree of downsampling increased, robustness trajectories increasingly diverged from the original metaweb, but the direction of this divergence depended on both the ecological mechanism used to remove interactions and the extinction scenario considered (Panel A; Figure 3).

[Figure 3 about here.]

Under body-mass ordered extinction, reductions in connectance produced increasingly divergent robustness responses relative to the metaweb, while trajectories relative to the niche and ATN reference states were weaker and more variable. This pattern was reflected in a strong interaction between connectance reduction and reference state ($\chi^2 = 384$, $p < 0.001$). Similar reference-dependent trajectories occurred under degree-based extinction ($\Delta_{connectance}$ × reference × downsampling approach: $\chi^2 = 98.7$, $p < 0.001$) and random consumer removal ($\chi^2 = 22.1$, $p = 0.001$), whereas random basal removal showed little evidence that the mechanism of downsampling altered robustness trajectories ($\chi^2 = 0.54$, $p = 0.997$)

Interestingly, burn-in further modified the relationship between downsampled networks and realised reference states. Comparison of robustness before and after burn-in showed that the process of dynamic modelling altered the magnitude and direction of divergence between downsampled networks and the metaweb, ATN, and niche references (Panel B Figure 3). Thus, the ecological assumptions imposed during downsampling do not fully determine the final properties of a realised network; subsequent dynamical processes provide an additional filter that can either reinforce or reduce differences among reconstruction approaches.

Together, these results demonstrate that reducing the connectance of a metaweb alone does not determine how inferred robustness changes; rather, the ecological assumptions governing which interactions are retained determine which structural properties of the metaweb are preserved and which are lost. As more links are removed the downsampling approach increasingly imposes these assumptions, different downsampling approaches produce distinct robustness trajectories, with the magnitude of these differences depending on the extinction scenario considered. Furthermore, dynamic burn-in modifies these relationships, demonstrating that the robustness of a reconstructed network at creation does not necessarily represent the robustness of the dynamically realised community. Therefore, conclusions drawn from extinction simulations depend not only on the extinction mechanism itself, but also on the ecological assumptions used to translate potential interactions within a metaweb into realised network structure.

Dynamic extinctions re-enforce ecological assumptions

To evaluate how interaction filtering influences dynamic behavior, we calculated pairwise contrasts of inferred robustness (R_{50}) across all combinations of community, extinction mechanism, and network type. In aggregate, topological extinctions (evaluated both at network creation and after dynamic burn-in (realisation)) yielded consistent robustness estimates across all empirically derived networks (Figure 4, left column). This structural cohesion was further reflected in hierarchical clustering, where empirically derived networks consistently formed a distinct cluster separate from the two synthetic benchmark networks (niche model and ATN; Figure 4, right column).

[Figure 4 about here.]

However, under dynamic extinction simulations, this grouping broke down. Introducing dynamic biomass interactions caused a distinct reshuffling in the relative relationships among network types (Figure 4, bottom row). Most notably, the downsampled niche networks yielded robustness estimates that converged with those of the synthetic niche model, whereas the remaining empirically derived networks maintained their mutual similarity. Additionally, the ATN model shifted to become more similar in robustness to the empirical cluster.

When examining topological extinctions, it is clear why empirically derived networks behave similarly to one another - they share a common baseline architecture inherited from the regional metaweb (Figure 1, panel B), and topological cascades depend strictly on network topology. For dynamic extinctions, however, network behaviour is not determined solely by initial structure. Instead, higher-order differences emerge from how functional parameterisation interacts with network topology. In the bio-energetic food web (BEFW) framework, species body mass dictates key functional parameters, including metabolic and consumption rates (Delmas et al., 2017). Because species body masses were assigned based on trophic position, the dynamic convergence of the synthetic niche model and the SVD-downsampled niche network likely reflects their shared trophic organisation. Consequently, dynamic simulations do not merely evaluate network structure; they actively re-enforce the ecological assumptions embedded during network construction.

6 Conclusions

Reconstructing local food webs from regional metawebs requires explicit assumptions about how ecological processes filter potential interactions into realised local networks. By comparing four downsampling approaches across topological and dynamic extinction scenarios, we demonstrate that the mechanism of interaction removal influences both the structural topology of the resulting networks and their inferred robustness to extinction.

6.1 Structural vs. dynamic fragility

Our results confirm that the qualitative discrepancy between topological and dynamic robustness identified by Curtsdotter et al. (2011) extends across alternative network reconstructions. Under most extinction scenarios, topological simulations inferred greater robustness than dynamic simulations regardless of downsampling approach. Topological extinctions treat networks as static topological dependency graphs, whereas dynamic extinctions incorporate biomass drawdown, competitive release, and rate-dependent secondary collapses. Consequently, static metrics consistently provide a conservative estimate of community vulnerability.

Importantly, while downsampling methods generated distinct initial topologies, their effects on topological robustness were relatively modest—likely because initial metawebs were already relatively sparse (mas Co = 0.20). However, once bio-energetic dynamics were introduced, subtle structural differences translated into substantial variations in dynamic persistence. Dynamic extinctions are strongly driven by node-level properties and metabolic rates rather than link topology alone. Thus, inferred dynamic robustness reflects an interaction between network structure and the physiological constraints imposed upon it.

Mapping empirical networks or model philosophy?

A central question arising from our findings is whether dynamic simulations evaluate empirical reality or simply mirror the assumptions of the underlying models (Strydom, Karapunar, et al., 2026). The dynamic convergence between SVD-downsampled metawebs and synthetic niche models highlights a fundamental circularity - because frameworks like the BEFW rely heavily on body-mass ratios and trophic positioning to derive interaction rates, networks constructed around niche-ordering assumptions naturally align with these dynamic engines. Conversely, downsampling methods that do not enforce trophic hierarchy (such as degree-product or random link removal) introduce structural configurations that interact differently with allometric scaling rules, leading to divergent dynamic behaviors.

This distinction is particularly evident when comparing the ATN and empirical downsampled networks. Although the ATN model is widely used as a realistic benchmark, its dynamic behaviour diverged from both the niche model. This underscores that no single synthetic network model can be treated as an absolute benchmark for realised food-web dynamics.

6.2 Implications for network reconstruction

While there is no single correct downsampling approach, our results offer practical considerations for empirically-derived metaweb downsampling:

- Alignment with dynamic frameworks: If the objective is to parameterise dynamic biomass simulations using allometric or bio-energetic models, SVD-based niche downsampling provides an internally consistent approach that preserves trophic hierarchy without imposing arbitrary degree distributions.
- Caution with unconstrained null models: Random downsampling introduces topological distortions that increase structural displacement during burn-in (Figure 1) and generate unpredictable dynamic responses, making it unsuitable as an ecologically informative filter.
- Evaluating static and dynamic stability together: Because topological extinctions represent an upper bound of structural robustness and dynamic extinctions reflect rate-dependent persistence, reporting both provides a clearer picture of network robustness.

Ultimately, downsampling a metaweb is an act of modelling ecological realisation rather than simple network reduction. Acknowledging the assumptions embedded within these downsampling filters is essential for accurately inferring community robustness in both ancient and modern ecosystems.

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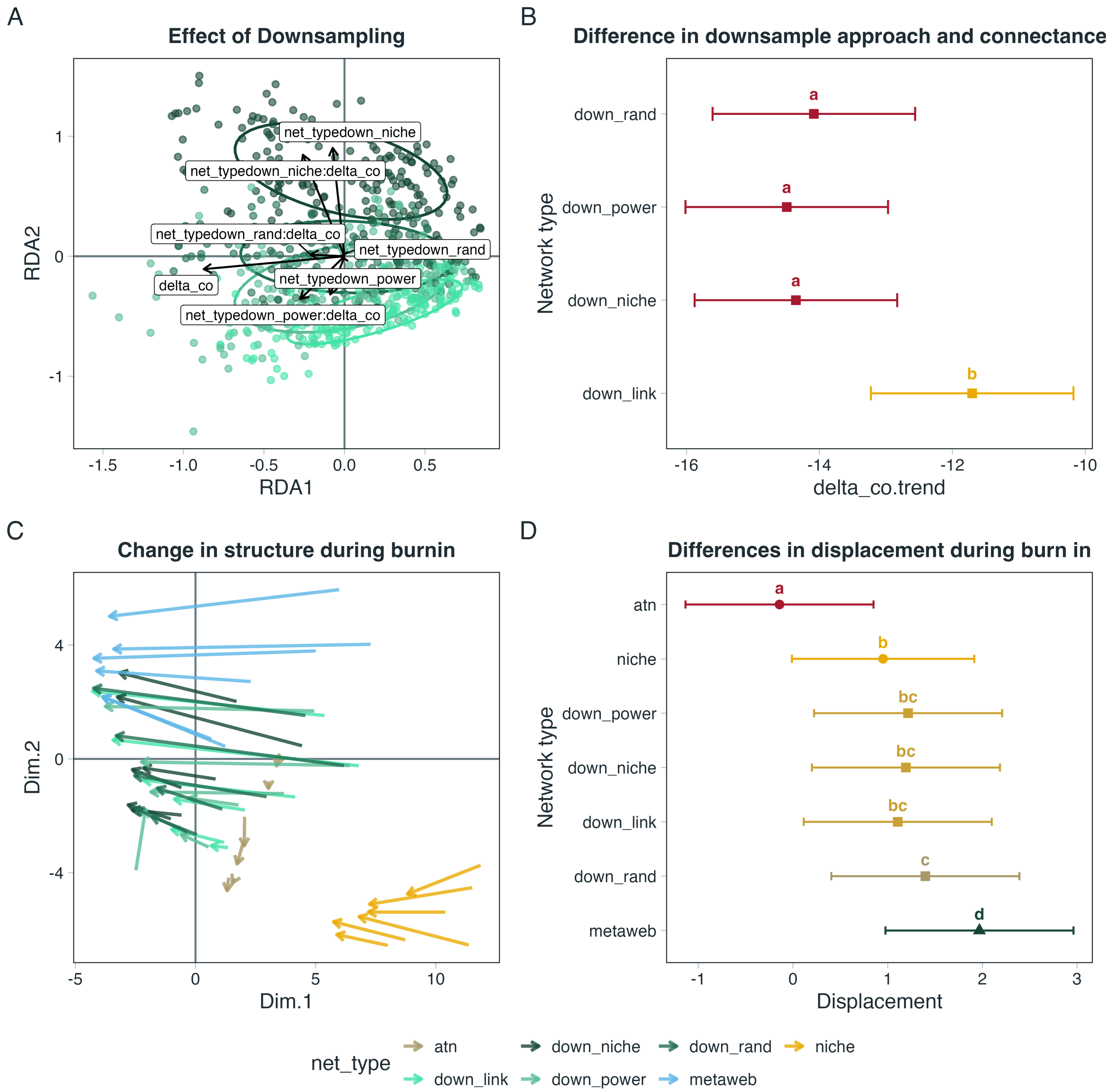

Figure 1: **Downsampling approach influences both network topology and subsequent structural reorganisation during burn-in.** (A) Redundancy analysis (RDA) ordination showing differences in network topology among downsampling approaches across the connectance gradient after accounting for community-level variation. (B) Estimated slopes (±95% confidence intervals) describing the relationship between the primary RDA axis and connectance change for each downsampling approach; letters denote post-hoc groupings from estimated marginal trend comparisons. (C) Principal component trajectories illustrating structural change between network creation and burn-in for each downsampling approach. (D) Mean Euclidean displacement through PCA space during burn-in (±95% confidence intervals); letters indicate pairwise differences among network types from estimated marginal means. Networks sharing the same letter are not significantly different.

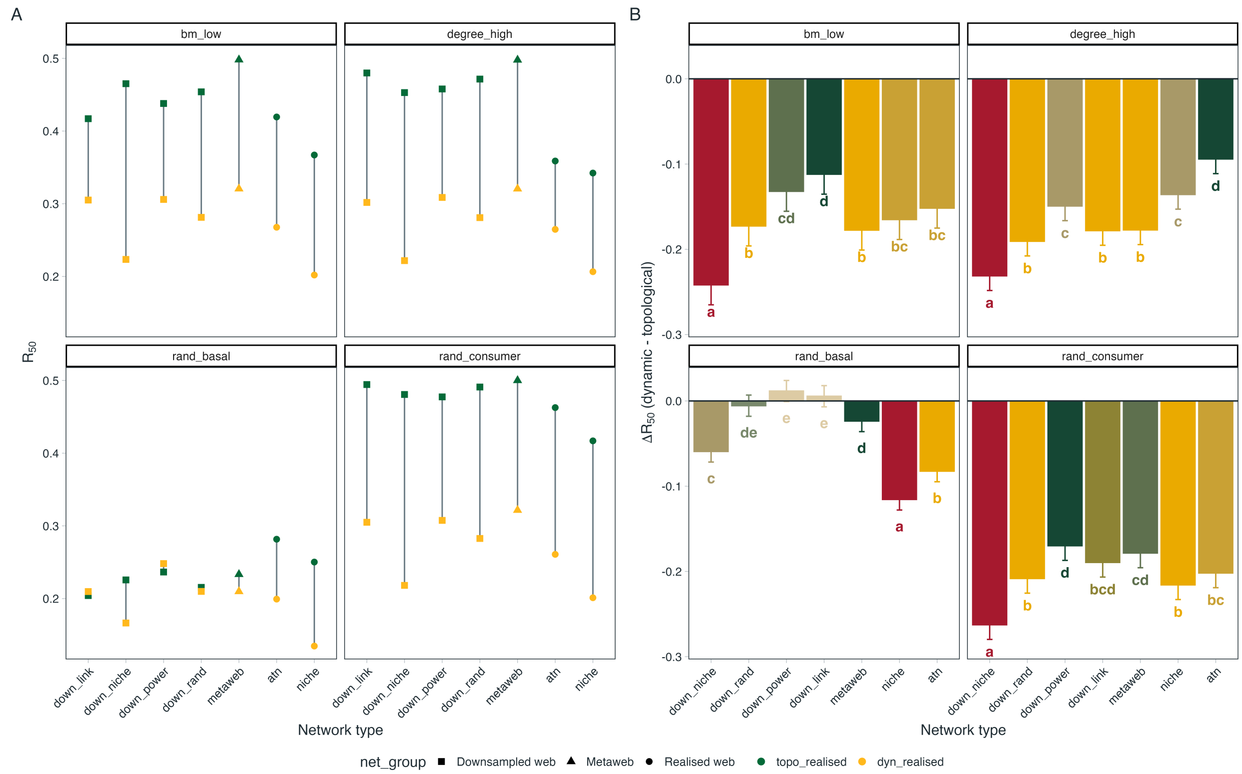

Figure 2: **Differences between topological and dynamic estimates of network robustness.** (A) Estimated robustness (R_50) of realised networks under topological (green) and dynamic (yellow) extinction simulations across network types and extinction scenarios. Points represent mean robustness values, with connecting lines highlighting the consistent reduction in robustness estimates when extinction cascades are evaluated dynamically. (B) Difference between dynamic and topological robustness estimates ($\Delta R_50 = R_{50,dynamic} - R_{50,topological}$) for each network type. Negative values indicate that dynamic simulations produce lower inferred robustness than topological simulations. Bars show estimated marginal means from linear models accounting for differences among empirical communities; letters indicate significant differences among network types based on Sidak-adjusted pairwise comparisons.

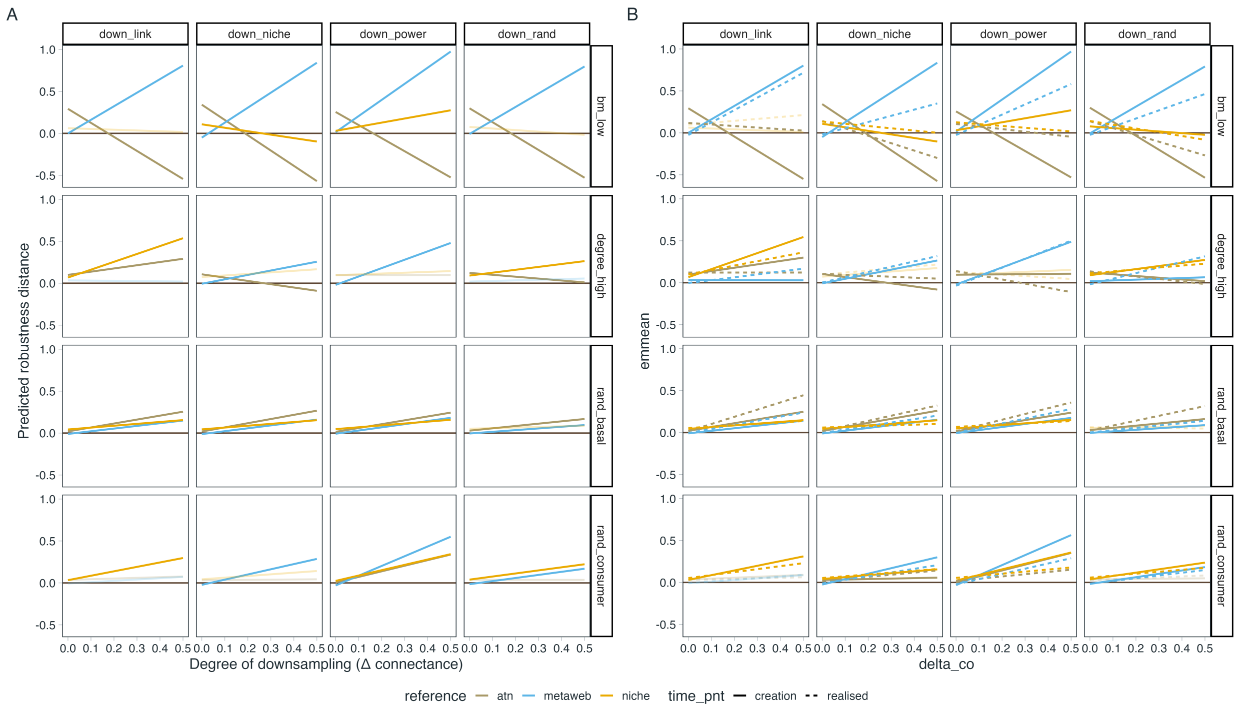

Figure 3: TODO.

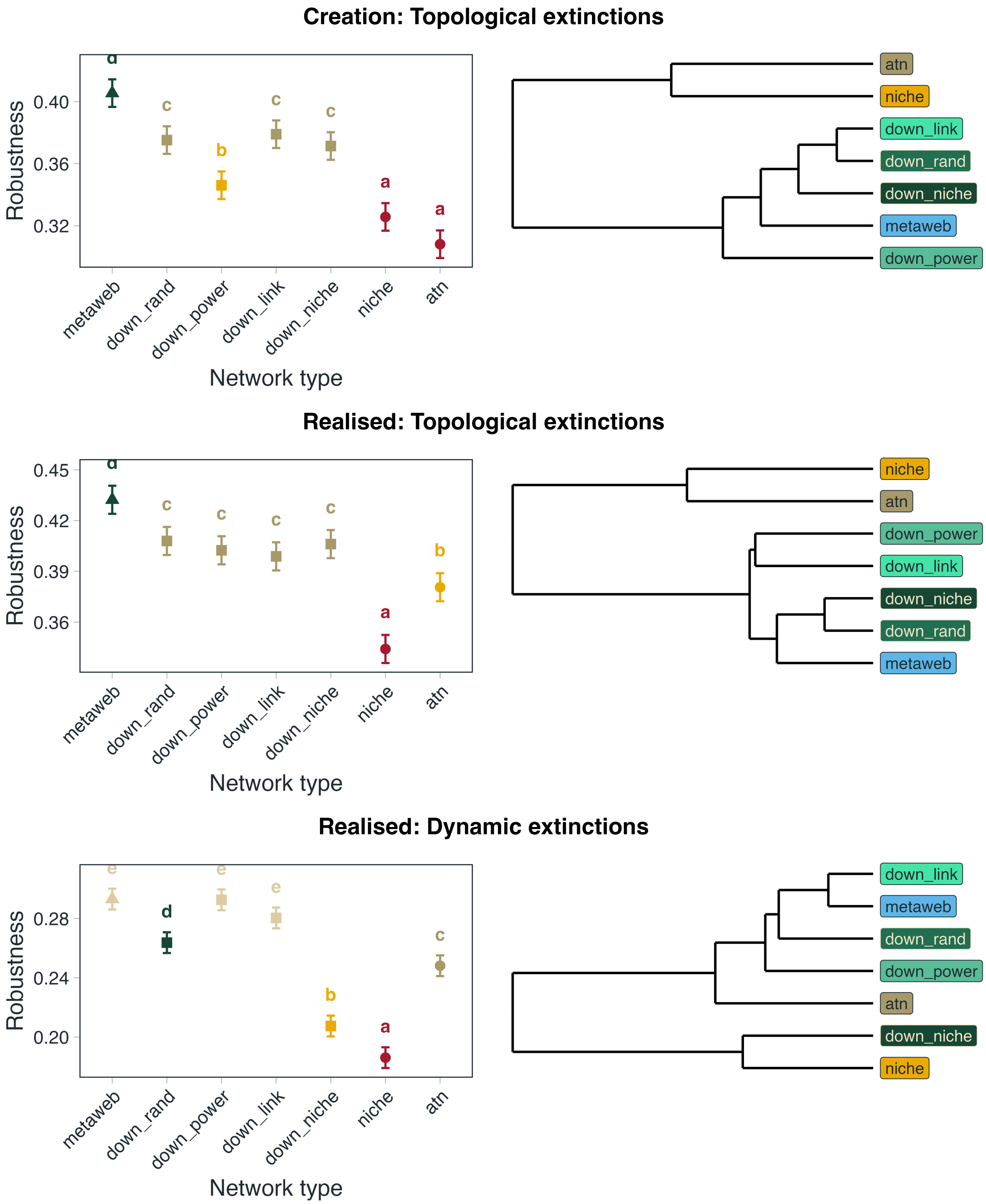

Figure 4: TODO